

Performance Work Statement (PWS)
Research and Development for Cost
Estimating Modeling
Data Analytics and Research (CEMDAR)
Development Effort
for

Air Force Cost Analysis Agency (AFCAA)

March 2026

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INTRODUCTION AND PURPOSE

The Department of the Air Force (DAF) has identified numerous gaps in the databases, methods, and tools it regularly uses for cost estimating and analysis and has committed to improving the efficiency and effectiveness of its cost estimating capabilities. This requirement invests in strengthening, accelerating and elevating cost data and analytic capabilities using the latest technologies to enable faster, more efficient, higher quality, and higher volumes of analytic decision support from the DAF cost community. It shifts the focus from desktop-based tools to cloud-based platforms; from stand-alone commodity-based datasets to integrated and relational databases containing programmatic, cost, technical, and schedule information; from static spreadsheets/reports to dynamic analysis and models connected to data pipelines; and from low-volume cost data modeling to highly scalable data collection, curation, analysis and models.

The objectives include (1) improving the quality and amount of data collected and available to cost estimators, (2) building centralized/ enterprise relational and integrated cost, schedule, and technical databases using data science best practices, (3) building the capability to optimize cost estimators' use of the larger amounts of data collected for cost estimating, and (4) providing centrally available and authoritative datasets and to develop cost estimating relationships, schedule estimating relationships, technical estimating relationships, benchmark tables, dashboards/visualizations, cost models, and other tools that Air Force and Space Force cost analysts can use to develop detailed, robust, and accurate cost estimates. The goal is to not only improve the timeliness and quality of program life-cycle cost estimates for budgeting, programming, and acquisition decisions but to enhance the capability to perform quick turn high-level force design and alternative capability analysis as well as lower level program element trade-off analysis.

Thorough and current cost datasets and databases are the cornerstones to the Air Force and Space Force continuously improving the accuracy and consistency of its acquisition program cost estimates required by statute (e.g., Title 10, United States Code § 3222, 3225, 4251, 4252, 4253, 4271, and 4328) and regulation (e.g., DoD Instruction (DoDI) 5000.02, DoDI 5000.73, DoDI 5000.75, DoDI 5000.80, DoDI 5000.85, DoDI 5000.87, and Air Force Instruction (AFI) 65-508) in support of DAF and Department of War (DoW) acquisition decisions. Moreover, these integrated datasets and databases as well as the accompanying analysis should serve as the basis to what the DAF cost community provides during the Strategy, Planning, Programming, Budgeting, and Execution processes. These analytically based cost estimates must be robust enough to identify available budget trade-space and what the near-term risks are as decision-makers balance competing priorities as they aim to deliver critical out-year capabilities to pace the threat.

Building credible and defensible cost estimates to assess program affordability and support budget development starts with complete and accurate datasets, sound estimating methods, models, and enterprise tools. This effort addresses focuses on primarily DAF requirements, but the solutions will be made available to benefit the government cost community, including the Office of the Secretary of Defense (OSD) Cost Assessment and Program Evaluation (CAPE) and other service cost agencies, and help it meet its growing demands.

TASKS AND TASK REQUIREMENTS

The contractor will develop enterprise datasets, methods and tools for use by cost estimators to meet the AFCAA Cost Estimating Modeling Data Analytics and Research (CEMDAR) objectives of an up-to-date, searchable, user-friendly, centrally available data and method repository that enables a high volume and timely delivery of quality, effective cost estimates. The contractor shall adhere to all scheduled timelines, milestones, delivery schedules, and administrative requirements to include scheduled on-time delivery of reports, data products, billing invoices, staffing of personnel, and action item completion. The contractor shall provide the necessary labor, supplies, equipment, and supervision to fulfill this effort as described in the following tasks and subtasks as outlined below.

The contractor will systematically and continuously document DAF and joint service cost estimating data requirements, then collect and store the required data, and provide analyses, methods, tools, applications, and models for use by cost estimators. Data will be collected in a manner that ensures complete and accurate documentation and based on the government's approval, as well as shared within the AFCAA's cloud-based platform(s) of choice (an appropriately designated platform/system for these products). Currently, systems utilized are withing the DAF Data Fabric and include DAF Data Visible, Accessible, Understandable, Linked and Trusted (VAULT) data platform and Envision platform.

The contractor shall be responsible for ensuring the accuracy, timeliness, and completion of all tasks under this effort, Replacement or substitute personnel for vacancies shall be filled with qualified personnel within fourteen (14) business days of vacancy, in accordance with *Table 1: Performance Thresholds* in Section IV, unless approved in writing or otherwise directed by the Contracting Officer (CO).

Meetings will include a kickoff meeting within ten (10) business days after the start of the contract award and each exercised option; meetings as needed between the contractor's Program Manager and AFCAA's Program Manager to track performance, resolve issues, and ensure compliance with the Performance Work Statement (PWS); status meetings as needed between the contractor's project leads and the AFCAA technical leads; and quarterly formal reviews to include a final review about 30 calendar days before the end of the period of performance between the contractor and the entire AFCAA government team to include the AFCAA Technical Director, AFCAA Program Manager, and the AFCAA technical leads. All meetings shall use Microsoft Teams as the default choice. Delivering the agenda and recapping the meeting, as outlined in *Table 2: Task Order Deliverables* in Section V, is the contractor's responsibility for the kickoff meeting and formal reviews, including the final review.

The contractor shall provide the COR with all passwords, data collected, processes used, and any other information obtained throughout the contract's period of performance. The COR will meet with the contractor within 7 business days prior to the end of the contract to ensure all information is received. Once confirmed, the contractor will send a letter to the COR confirming that the data rights requirements (paragraph 0) were completed not later than five business days prior to the end of the contract.

Task 1 (Enabling task) – Obtaining Permissions to Access Data Sources

Successful performance of this requirement depends on the contractor working proactively with the government to obtain permissions to access government held data. Much of the data in the Cost Assessment Data Enterprise (CADE) repository is from defense contractors and major subcontractors. Each considers its data as business sensitive and subsequently has marked it, or considered it, as proprietary. This requires protecting this information from public disclosure. Since the contractor team while providing these services will be provided or have access to nonpublic information, each person on the contractor and subcontractor team working this effort shall sign a non-disclosure agreements (NDA) with AFCAA (**Error! Reference source not found.**) and an NDA with OSD CAPE for CADE (**Error! Reference source not found.**). The contractor and subcontractor team shall work proactively and sign, as required, NDAs with other government organizations to support and fulfill data collection and task requirements.

The contractor will work proactively with AFCAA to identify NDA requirements for all Original Equipment Manufacturer (OEM) and major subcontractors. Should an OEM or major subcontractor require an NDA with the contractor, then the contractor shall develop and complete this before AFCAA provides access to this OEM's or major subcontractor's government-held data. These NDAs must carry through the entire contract period of performance, including all option contracts. Furthermore, the NDAs should flow equally across the contractor and its subcontractors.

As part of its proposal and throughout the duration of this contract, the contractor shall identify and notify per paragraph **Error! Reference source not found.** the Contracting Officer of any Organizational Conflict of Interest (OCI) it or any of its subcontractors has, or could have, with any OEM or major subcontractor who has submitted reports held in CADE or the Earned Value Management Central Repository (EVMCR). Moreover, the contractor shall notify the Contracting Officer and the AFCAA Program Manager of any acquisition or merger involving itself, or its subcontractors and explain whether that action will cause create a OCI. Similarly, the contractor will identify any other emerging event or action that would create a OCI.

Task 2 (Enabling task) – Develop a Plan of Action and Milestones (POA&M)

The contractor shall develop and submit, for government review and approval, a POA&M describing the contractor's approach to satisfy all tasks and elements for data ingestion and curation, analysis, and other task requirements for the base period within 20 business days of contract award and for each option year within 15 business days of the option year start. The POA&M will be a living document that will be modified throughout the entire contract period of performance, including all option periods.

- a. As part of the kickoff meeting and at the start of the base period and each option period, the contractor and AFCAA will meet to discuss POA&M development and subtasks for inclusion. A draft POA&M shall be delivered for review and comment not later than (NLT) 20 business days after contract performance begins (and within 15 business days after each option year begins). The contractor will work with the

- 123 AFCAA Technical Director, its Technical Leads, and its Program Manager to obtain
124 approval of a final POA&M. The POA&M will include, as a minimum:
- 125 b. A detailed description of the approach to completing each task (with element level
126 descriptions) and subtask.
 - 127 c. Task and subtask activity and milestone schedule, including interim, draft, and final
128 milestones, products, and contract project deliverables and a description of risk
129 associated with completing the planned schedule. Interim milestones should include
130 any events or progress points that would require a go or no-go decision from the
131 government based on research to-date prior to moving ahead. The schedule will be
132 used for tracking purposes and should be included in each Status Reports.
 - 133 d. The contractor's plan should include sufficient lead time for the delivery of all project
134 deliverables to allow the government to review and provide feedback. The contractor
135 shall resolve all the government's comments before any deliverable can be considered
136 final. This shall occur not later than ten business days prior to the end of the contract
137 base period and each contract option year end.
 - 138 e. Each task and subtask will document approvals to collect classified data. This
139 includes tracking potential data aggregation or compilation as determined by program
140 classification guides. The contractor shall ensure the collected data are protected at
141 the appropriate level necessary to safeguard this information. The contractor shall
142 work closely with the AFCAA technical leads to gain approval to collect classified
143 data.
 - 144 f. A management plan explaining the contractor's approach to staffing, schedule
145 management, business management, and reporting. This plan will track and align to
146 both task and subtask activity and schedule. The plan will be specific in the Chief
147 Cost Data Scientist and Program Manager (PM) role and activities.
 - 148 g. Plans to format and share CEMDAR deliverables as working documents and to
149 finalize all products for upload onto AFCAA's cloud-based platform(s) of choice (for
150 example, currently the DAF VAULT data platform), the CADE system, or any other
151 designated repository. These deliverables will include, but are not limited to, all
152 datasets, databases, tools, models, reports, documents, manuals, Data Item
153 Descriptions, training materials, and dictionaries required to upload and share the
154 CEMDAR products.

155 The contractor shall be available to brief the POA&M to the government, along with any risk
156 areas associated with completing the planned schedule, to the AFCAA Technical Director, the
157 Program Manager, and the COR. The POA&M is a 'living document' and the contractor will
158 provide recommended changes/updates to the government at the quarterly meetings, or as
159 needed, and will be given direction and/or approvals by the government on the recommended
160 changes/updates to this document.

161 After the government sends its statement of intent to exercise its next option, AFCAA will
162 inform contractor of the specific requirements to be incorporated into the POA&M. The
163 contractor will respond to AFCAA's requests to responds and update the POA&M.

Task 3 (Core task) – Development and Continuous Improvement of Data Pipelines for Enterprise Cost and Technical Analysis

Task Objective

The contractor shall design, develop, and implement a series of robust, automated data pipelines to integrate disparate data sources into a unified data environment, enabling enhanced cost, schedule, and technical analysis. These pipelines will conform to the organization's defined data operating model and support recurring and ad-hoc analytical needs. The pipelines will be developed in the AFCAA's platform of choice, optimally utilizing available DAF technologies/toolsets. Currently, systems utilized are within the DAF Data Fabric and include DAF Data Visible, Accessible, Understandable, Linked and Trusted (VAULT) data platform and Envision platform, but this is subject to change over time as the DAF Data Fabric evolves. The ultimate goal is to leverage these data pipelines so that they provide reliable, accurate, and readily accessible data to improve cost estimating methods and other analyses, aligning with the Cost Analysis Data Operating Model (CADOM)

Scope of Work

The contractor shall perform the following subtasks, as required by AFCAA in the approved POA&M:

Cost Analysis Data Operating Model (CADOM) Data Architecture Design/Support

- Collaborate with government stakeholders to assist in data architecture design/support as required, to include:
 - Assistance in conducting assessment of platforms, toolsets and assistance in defining the optimal solution/tech stack meeting all AFCAA's requirements and considering all constraints. This also includes ensuring innovative methods are utilized throughout the development of the CADOM
 - Assistance in architecting solutions that meet unique government requirements, such as data and applications that span various classification environments, and spanning users that have different levels of access to data
 - Assistance in maintaining the Standard Operating Procedures (SOPs) for the CADOM
 - Assistance in developing and incorporating optimal process user/Subject Matter Expertise (SME) feedback and workflows

Data Source Identification and Assessment:

- Collaborate with government stakeholders to identify and document relevant data sources across the organization to feed the data pipelines described below. Note that data and data pipelines are expected to include (but are not limited to): Cost and

Schedule Data Reporting (CSDR), Earned Value Management (EVM), Selected Acquisition Report (SAR/MSAR), Contracts, Budget, Air Force Total Ownership Cost (AFTOC), etc. AFCAA will work with pipeline owners to allow access as needed to the contractor team.

- Conduct a comprehensive assessment of each data source, focusing on:
 - **Data Quality:** Evaluate data accuracy, completeness, consistency, and fidelity using defined metrics (e.g., leveraging a Data Fidelity Scorecard like the example in reference g of this document).
 - **Data Normalization Requirements:** Determine the necessary normalization processes, including inflation/price escalation adjustments, Work Breakdown Structure (WBS) alignment, and Cost Element Structure (CES) standardization. Justify the chosen normalization methods.
 - **Data Availability and Access:** Investigate data security classifications, aggregation limitations, proprietary restrictions, access requirements, and control information. Adhere to all applicable security policies and regulations.
 - **Data Source Characteristics:** Document any other relevant database characteristics (e.g., data formats, API availability, data update frequency) that may affect data collection, management, and use.

Data Pipeline Design and Development:

- Design data pipelines that adhere to modern data architecture principles and leverage appropriate technologies (e.g., Apache Spark, Python, R, SQL, Databricks, cloud-based data warehousing solutions).
- Develop modular, scalable, and maintainable data pipelines for data extraction, transformation, and loading (ETL) processes.
- Implement automated data quality checks and validation rules wherever possible and optimal within the pipelines to ensure data accuracy and consistency.
- Develop error handling and logging mechanisms to facilitate pipeline monitoring and troubleshooting.
- Development/maintenance of a “metadata” data pipeline/layer in order to connect the distinct and disparate datasets with an overarching framework of metadata architecture/fields. This metadata layer shall connect into all the data pipelines to the greatest extent possible such that the user can query fields from the metadata layer and return data at various levels of detail for each of the related data pipelines.

Data Ingestion and Transformation:

- Ingest data from identified sources into a raw data storage area (e.g., a data lake).
- Perform data cleansing, normalization, and transformation operations to conform to the organization's data model.

- Create curated data tables and views optimized for analytical queries and reporting.
- Implement data lineage tracking to maintain traceability from raw data to transformed data.

Data Repository Development and Maintenance:

- Organize data tables and views into user-friendly, searchable data repositories (such as a data catalog), leveraging existing available technologies wherever available, and customizing solutions where available technologies are not present in the cloud-based platform of choice.
- Ensure the data repositories are optimized for performance and scalability.
- Implement data security measures to protect sensitive data.
- Develop and maintain data dictionaries and metadata documentation to facilitate data understanding and usage.
- Document and maintain data fidelity information (e.g. the data fidelity scorecard example in reference g) to facilitate user data understanding and usage

Documentation and Training:

- Develop comprehensive and user-friendly documentation for each data pipeline, including data source descriptions, ETL processes, data transformations, data quality checks, and data lineage information. Integrate, wherever possible the documentation for data pipelines within the CADOM architecture in an efficient way (i.e. GitHub, markdown, tooltips, Wiki, etc).
- Conduct training sessions for users to demonstrate the use of the data repositories and analytical tools.
- Provide detailed documentation for future enhancements and maintainers of the developed pipelines.

Continuous Improvement (Once data pipelines have been developed and deployed):

- Monitor data pipeline performance and stay abreast of new technologies and best practices in data engineering identify areas for optimization, suggesting potential improvements to AFCAA, where applicable.
- Implement regular maintenance and updates to ensure data accuracy and availability.
- Respond to user feedback on any issues with the pipeline or products within CADOM and address these issues as well as data quality issues promptly.

AFCAA will evolve and adapt its Cost Analysis Data Operating Model (CADOM) over time to adapt to changes/evolutions in available platforms, technologies and toolsets. The current CADOM architecture is pictured in Figure 1 for reference.

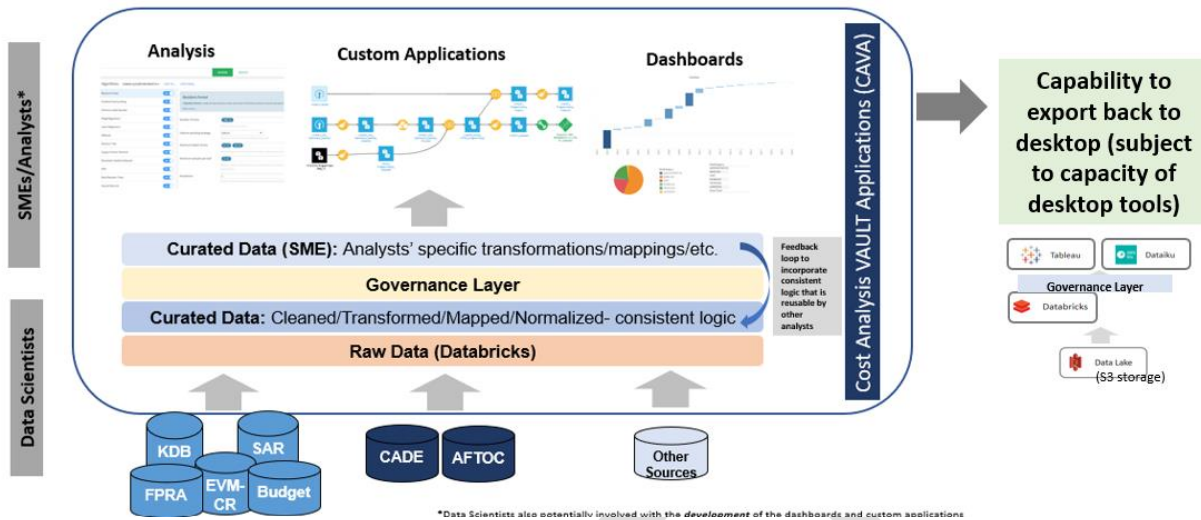


Figure 1- Current Cost Analysis Data Operating Model (CADOM) Architecture, for reference purposes

Data Fidelity Scorecard. The contractor shall support AFCAA in collaborations with other cost organizations to identify data sources for collection efforts. These include but are not limited to: the Air Force Life Cycle Management Center, the Space Systems Command, the Air Force Nuclear Weapons Center, the Air Force Sustainment Centers, the Air Force Rapid Sustainment Office, the Naval Cost Agency, the Naval Systems Commands, the Deputy Assistant Secretary of the Army for Cost & Economics, the OSD CAPE, the Department of Homeland Security, the National Aeronautics and Space Administration, the National Reconnaissance Office, the Navy Strategic Systems Program Office, and the Missile Defense Agency.

Task 4 (Core task) – Data-Driven Analysis/Studies, Model Development and Application Deployment

Data Fidelity Scorecard Task Objective:

The contractor shall perform in-depth cost, schedule, and technical analyses and/or studies, develop dynamic predictive models, custom applications, and insightful visualizations; and deliver actionable recommendations to improve cost estimating methodologies and decision-making across the organization, as required by the approved POA&M. The goal of this task is to provide data-driven insights, analysis, and tools tailored to AFCAA's needs. This task will include projects that are intended to be fully deployed within the CADOM, projects that are intended to be fully deployed outside the scope of the CADOM, and those that are a hybrid of these two. This task shall ensure optimal delivery and integration of analysis/studies within the CADOM ecosystem or as standalone deliverables, as determined by AFCAA. The contractor shall leverage, to the greatest extent possible, the existing and future data pipelines developed and delivered in Task 3 for Task 4 projects. The contractor shall assist the government in

identification of opportunities to migrate traditional studies/analysis to the CADOM and/or other modern platforms/toolsets wherever synergies and/or efficiencies could be achieved.

Scope of Work:

The contractor shall develop use cases, dynamic analysis/predictive models, custom applications, and conduct cost and technical studies as required per approved subtasks in the POA&M. The subtasks in the POA&M, depending on the type of task, will include some combination of data analysis, model development, application/tool development, and conducting targeted cost and/or technical studies. Subtasks as defined in the POA&M could fall into one or more of these categories as listed below. Each category is further defined below, along with potential expectations and/or examples of what will be required (each subtask will be clearly defined within POA&M). To facilitate planning and illustrate the scope of work under this task, a representative list of example projects is provided in Appendix A. These projects are indicative of the types of data-driven analyses, studies, and model development efforts that will be executed annually for the base year and each subsequent option year. The Government will introduce new projects each year. It is understood that project timelines will vary; some larger projects may span several years, while others may be completed in less than a year. Furthermore, it is acknowledged that for many of the studies undertaken, readily available data may not exist, and data collection and normalization will be a key component of the research effort.

Data Analysis, Model Development, Cost and/or Technical Studies:

- Conduct thorough data analysis and/or develop models using available data pipelines and/or other data sources.
- Identify trends, patterns, and relationships within the data relevant to the identified use cases, including but not limited to cost estimating benchmarks, methods, Cost Estimating Relationships (CER), Schedule Estimating Relationships (SER), Technical Estimating Relationships (TER), and other analysis/estimating approaches, as required.
- Develop end-to-end cost model architectures and/or models, that could be used as a template/ starting point, early acquisition trades, etc.
- Employ appropriate statistical and machine learning techniques to build predictive models that improve forecasting accuracy and risk assessment.
- Ensure all models are rigorously tested and validated using appropriate statistical methods.
- Conduct targeted studies to meet mission-unique requirements for specific commodities and/or program.
- Create interactive dashboards and visualizations that provide insights into key performance indicators and trends.

340 *Application and Tool Development:*

- 341 • Design and develop custom applications and tools to support cost, schedule and
- 342 technical estimating/ analysis. This could include, for example, development of cost
- 343 and schedule toolsets/applications that could then be leveraged by analysts on specific
- 344 program's estimates.
- 345 • Develop user-friendly interfaces that enable stakeholders to easily access and utilize
- 346 the developed tools and applications.

347 *CADOM Deployment (as required by approved POA&M):*

- 348 • For projects intended for integration/deployment within the CADOM cloud-based
- 349 platform(s), the contractor shall work with AFCAA to ensure compliance with
- 350 CADOM architecture, data standards, and security requirements.
- 351 • Deploy developed models, applications, and tools to the CADOM environment
- 352 following established SOPs.
- 353 • Provide any troubleshooting and/or continuous improvement for deployed
- 354 applications within the CADOM as required
- 355

356 *Final Product/Report/ Documentation:*

- 357 • To the greatest extent possible, the projects, use cases, applications, dashboards, etc
- 358 developed within CADOM should be deployed within the cloud-based platform(s) of
- 359 the government's choice. They should also leverage integrated documentation native
- 360 within the platform(s). This could be, for example, code documentation in GitHub,
- 361 markdown, wikis and/or other existing and custom tool tips/interactive documentation
- 362 wherever optimal and appropriate.
- 363 • Certain projects will involve scope both inside and outside of the CADOM. In this
- 364 case, a hybrid approach could include both deploying models and applications to the
- 365 CADOM platform, and development of a traditional report/documentation could still
- 366 be required as well.
- 367 • For projects outside the scope of the CADOM, or hybrid projects that documentation
- 368 within CADOM is not feasible, the contractor shall develop comprehensive reports
- 369 that document the analysis, methodology, findings, and recommendations.
- 370 • Reports will include the following sections: Purpose, Background, Scope and
- 371 Limitations, Ground Rules and Assumptions, Methodology (data, normalization
- 372 process, normalized data, and analytic methods), Risk/Uncertainty,
- 373 Results/Conclusions/Recommendations.
- 374 • Data sources used to populate the data collection will be clearly identified and
- 375 provided as reference material, to include the source data extracted and used when
- 376 required.
- 377 • All normalization and analysis must be fully documented in detail so that the analysis
- 378 can be recreated and normalized data is fully traceable back to the raw data.

- All instructions/training materials must be fully documented for repositories/databases/tools developed. Source data and normalized data will be provided in the final deliverable report or delivered dataset/database/tools.

Training and Knowledge Transfer:

- Develop and deliver training materials and workshops to educate users on the use of the developed tools, applications, and models.
- Provide deployment training to users and collect feedback to refine the original training materials
- Provide ongoing support to users and address any questions or issues that arise.
- Document all development activities and share knowledge with government stakeholders to ensure long-term sustainability.

Task 5 (Core task) – Contracts Database (Option CLIN)

The contractor shall update an existing AFCAA Contracts Database with additional contract information for programs that already exist in the database as well as add new programs to the database. The contractor shall ensure the fields of the Contracts Database, as it exists in CADOM cloud-based architecture (currently, VAULT), are relatable to metadata from datasets, such as CSDR/FlexFile, SRDR, SAR, and others as identified by AFCAA, ingested per paragraph

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AFCAA will identify new priorities for the Contracts Database at the start of each option period. In addition, the contractor shall perform the following tasks:

- a. Review lessons learned and business rules for contract growth buckets and redesign features/mapping as necessary for database.
- b. Incorporate new features per user input and incorporate data fixes for new features not backward compatible yet. Leverage business intelligence (BI) tools for data ingestion, normalization, and visualizations, as appropriate.
- c. Update user guides for each version of the KDB tools to include new features or other incorporated changes.
- d. Add new data to the database. Make corrections as needed to data records in the database.
- e. Provide database updates on a quarterly basis. Ensure delivered versions of the KDB database and associated tools/documents are available to all DoW CADE users, via platform designated by government
- f. Update training material and provide one combination virtual and in-person (National Capital Region) course to KDB users. Sub-tasks include:
 - i. Updating existing training material to reflect current KDB structure, content, data tagging, etc.
 - ii. Applying the data to cost estimating use cases to include products associated with Task 4 or projects assigned when directed under this task (e.g.,

- engineering change order analysis from data). **Error! Reference source not found.**
- iii. Providing a demonstration of the KDB tools and major database updates with an emphasis on how a cost estimator can use these tools.

Task 6 (Core task) – DAF CSDR Support (O&M funded, Option CLIN)

The DoW requires accurate and reliable cost reporting to inform life cycle cost estimates and programming and acquisition decisions. The data collected from DoW cost reporting forms the basis of these estimates, as mandated by statute and DoW and DAF policies.

The contractor shall provide support to the Air Force Cost Analysis Agency (AFCAA) and the DAF Cost Centers to administer the Cost and Software Data Reporting (CSDR) process for ACAT I level DAF programs and all programs delegated to AFCAA for CSDR plan approval, in accordance with DoDI 5000.73, Cost Analysis Guidance & Procedures. This support would include:

- a. CSDR Planning, Verification and Validation, and Compliance: The contractor shall assist AFCAA, Center Cost Staffs, and the Program Management Office (PMO) in all aspects of planning, validation, and compliance as requested by AFCAA, for example:
 - i. Ensuring timely identification of upcoming solicitations and request for proposal (RFP) releases as well as tracking current submissions against due dates, etc
 - ii. Ensuring that approved CSDR plans represent the broader cost community data needs for future cost estimating.
 - iii. Ensuring that valid data is submitted as required by the final approved CSD by comparing submissions to plans and prior submissions
- b. CRSB Support: The contractor shall work with AFCAA, Center Cost Staffs, and the Program Management Office (PMO) to identify issues for the Cost Reporting Standards Board (CRSB) and working with the AFCAA CSDR Process Manager on CRSB recommendations.
- c. CSDR Process Management: The contractor shall support the AFCAA CSDR Process Manager in management of the entire end to end process of CSDR planning, verification and validation, and compliance for DAF program CSDRs. Note that all actual report approvals will only be accepted by government personnel, but the contractor will function in a support role by performing verification and validation analysis. This will require interaction and collaboration with the support contractor and government staffs of the Office of the Secretary of Defense (OSD) Defense Cost and Resource Center (DCARC) office for DAF ACAT I level programs, and the DAF Cost and Resource Center (DAFCARC) team for DAF ACAT II-III level programs.
- d. Cost Working Integrated Product Team (CWIPT) Support: The contractor shall support the CWIPT for DAF ACAT I level and ACAT II-III level programs, assisting in the development of a final recommended CSDR plan for approval.

Task 7 – Surge Support (RDT&E, Option CLIN)

During the performance period of this PWS, the Government may require services that expand beyond the ongoing support anticipated. This optional surge support CLIN provides a pre-negotiated mechanism for the Government to acquire additional expertise from cost estimators, data scientists, and technical subject matter experts. The specific tasks and roles covered by this surge support are defined within this PWS.

The Government may exercise this option CLIN at its discretion. The decision to invoke this task will be based on the required level of support, project duration, and the availability of project funding. Exercising this option allows the Government to quickly augment the team with pre-approved roles and labor rates to meet emergent mission requirements without the delay of a new procurement action.

Task 8 – DAF Surge Support (O&M, Option CLIN)

During performance identified in this PWS, it is possible that the effort could potentially expand past the anticipated, on-going DAF efforts for Support. This task refers to efforts that the government may exercise at its discretion to obtain cost estimators, data scientists, and technical subject matter experts for surge support to perform specific task descriptions and add these to this contract via this option CLIN for any additional requirements. The government will define specific task descriptions and add these to this contract via this option CLIN. The decision to invoke the task shall be based on the level of support that is required, project length, and the amount of available project funding.

Knowledge, Skills, and Abilities (KSA)

The contractor is responsible for ensuring its personnel (to include subcontractors) fully comply with the minimum experience and knowledge described in the following paragraphs... labor category... staff appropriate mix of skillsets

General Requirements

The contractor shall assign personnel to perform tasks outlined in this PWS. The contractor shall notify the PCO of any proposed changes to key personnel. Substitutions of key personnel require PCO approval. Also, the contractor can request from the AFCAA Program Manager and COR waiver considerations for years of experience should a candidate possess unique or hard to find skills. Administrative and clerical support under this contract is not considered a direct task or labor category.

Program Manager

The Program Manager manages the coordination and completion of tasks and subtasks/projects on time, within budget, and within scope. The Program Manager assigns responsibilities, leads and directs the work of others, works with project leads to set deadlines, and monitors project timelines. He or she prepares reports regarding the status of each project. A wide degree of creativity and latitude is expected from the Program Manager.

The Program Manager requires a Bachelor of Science or Bachelor of Arts degree from an accredited college or university and experience in the program management field or in a related area and familiarity with management concepts, practices, and procedures. This person must have a minimum of five (5) years managing projects of comparable scope to the CEM effort and a experience in managing similar cost estimating projects.

Chief Cost Data Scientist

The contractor shall provide an individual to serve as the Chief Cost Data Scientist. The Chief Cost Data Scientist is required to manage and oversee all CEMDAR contractor activities from a technical standpoint, including subcontractors, to satisfy the requirements identified in this PWS. The Chief Cost Data Scientist is expected to seek innovative and transformative methods to meet the objectives of this requirement. His or her responsibilities include ensuring technical content quality, commonality, and configuration compliance across all tasks.

The Chief Cost Data Scientist requires a Bachelor of Science or Bachelor of Arts degree from an accredited college or university and personnel in this category must have a minimum of three (3) years performing projects of comparable scope to the CEM effort, a minimum of eight (8) years as a data scientist, as well as five (5) years in cost estimating/analysis. The Chief Cost Data Scientist shall demonstrate the following:

- a. Leadership in Data Science – Experience in leading cross-functional teams of data scientists, data engineers, and data/cost analysts. Proven ability to define and articulate a data strategy aligned with organizational goals, identifying opportunities for leveraging data to drive business growth and innovation.
- b. Integration of Capabilities – Expertise in integrating advanced data science capabilities with existing processes and technologies, ensuring alignment and maximizing efficiency across departments.
- c. Stakeholder Engagement – Exceptional communication skills to engage with stakeholders, translating complex data insights into actionable strategies and recommendations for business decisions.
- d. Innovation – Demonstrated history of driving research and development initiatives in machine learning and Artificial Intelligence (AI), exploring cutting-edge technologies and methodologies to keep the organization at the forefront of data science advancements.
- e. Cost Estimating Experience – Knowledge and experience in performing cost estimating and analysis on major defense systems including typical work breakdown structures, standard cost data (e.g., CSDRs, O&S, EVM, SAR, Budget data), development of cost estimating relationships and analogy estimating, and basic cost estimating experience

Expert Cost Estimator/Analyst

The contractor shall provide expert cost estimators or cost analysts that have comprehensive and in-depth experience in cost analysis and mastery of Air Force and/or Space Force weapons

systems. The contractor shall provide expert cost estimators/analysts who can integrate seamlessly with and mentor other weapons system cost estimators/analysts. These expert cost estimators/analysts must have expertise in cost analysis principles including but not limited to:

- a. **Cost estimating** – Recognized expertise in DAF weapon systems, including innovative estimating methodologies that optimize data collection parameters. Demonstrated mastery of DAF weapon systems as well as system-specific programmatic and technical cost drivers associated with both acquisition and operating and support (O&S) phases of an acquisition program's life cycle. Proven ability to develop new cost estimating methodologies.
- b. **Data collection** – Mastery of collecting and normalizing cost, technical, and programmatic data from government entities. Extensive knowledge of the Defense Cost and Resource Center (DCARC) data collection process and deep experience in the use of and analysis of the Cost and Software Data Reports (CSDR) and Integrated Program Management Reports (IPMR)/Contract Performance Reports (CPR). Extensive knowledge of DoW VAMOSC Systems, e.g., AFTOC, Navy VAMOSC, OSMIS, EVAMOSC.
- c. **Leading teams** – Proven ability to lead and mentor cost research project teams to successful conclusions.
- d. **Data analysis and cost tools** – Mastery of data analysis and developing advanced cost estimating methods and tools. Can serve as a SME to others.
- e. **Experience** – Must have at least fifteen (15) years of cost analysis experience.

Senior Cost Estimator/Analyst

The contractor shall provide senior cost estimators or cost analysts that have significant experience in cost analysis and expertise with Air Force and/or Space Force weapons systems. The contractor shall provide senior cost estimators or analysts who can integrate with the other weapons system cost estimators or analysts. These senior cost estimators/analysts must have experience in cost analysis principles including but not limited to:

- a. **Cost estimating** – Demonstrated expertise in DAF weapon systems, specifically estimating methodologies that facilitate identification of data collection parameters. Includes an in-depth understanding of DAF weapon systems as well as system-specific programmatic and technical cost drivers associated with both acquisition and operating and support (O&S) phases of an acquisition program's life cycle.
- b. **Data collection** – Demonstrated expertise collecting and normalizing cost, technical, and programmatic data from government entities. Knowledge of the Defense Cost and Resource Center (DCARC) data collection process and demonstrated experience in the use of and analysis of the Cost and Software Data Reports (CSDR) and Integrated Program Management Reports (IPMR)/Contract Performance Reports (CPR). Knowledge of DoW VAMOSC Systems, e.g., AFTOC, Navy VAMOSC,

- OSMIS, EVAMOS. Leading teams – Demonstrated ability to lead cost research project teams to successful conclusions.
- c. Data analysis and cost tools – Demonstrated knowledge and demonstrated expertise/ability to analyze data and develop cost estimating methods and tool.
 - d. Experience – Must have at least ten (10) years of cost analysis experience.

The contractor shall ensure they have identified individuals in each one of the following categories for senior level cost and commodity specific Subject Matter Expertise (SME). One individual may qualify for multiple commodity areas. The individuals identified need to be either currently on staff and/or contractor must have the ability to onboard SMEs within the timeframe they are needed to support requirements, per the approved POA&M schedule. Senior Cost Analysts should have at minimum 10 years of experience in cost analysis as well as years of experience as identified each of the commodity areas. The requirement for individual commodity experience necessary is subject to vary each year, per approved POA&M.

Senior Cost Estimator/Analyst – Aircraft SME

Must have at minimum 10 years of experience in cost analysis as well as at least seven (7) years of progressive DoW experience, e.g., Air Force, Space Force, Navy, Army, MDA, and other DoW organizations. Preferably Air Force/ Space Force ACAT I threshold level aircraft acquisition program cost analysis experience.

Senior Cost Estimator/Analyst – Space and Launch Systems SME

Must have at minimum 10 years of experience in cost analysis as well as at least seven (7) years of progressive DoW experience, e.g., Air Force, Space Force, Navy, Army, MDA, and other DoW organizations. Preferably Air Force/ Space Force ACAT I threshold level acquisition program cost analysis experience.

Senior Cost Estimator/Analyst – Tactical Missiles SME

Must have at minimum 10 years of experience in cost analysis as well as at least seven (7) years of progressive DoW experience, e.g., Air Force, Space Force, Navy, Army, MDA, and other DoW organizations. Preferably Air Force/ Space Force ACAT I threshold level tactical missile/munitions acquisition program cost analysis experience.

Senior Cost Estimator/Analyst – Ballistic Missiles SME

Must have at minimum 10 years of experience in cost analysis as well as at least seven (7) years of progressive DoW experience, e.g., Air Force, Space Force, Navy, Army, MDA, and other DoW organizations. Preferably Air Force/ Space Force ACAT I threshold level ballistic missile/munitions acquisition program cost analysis experience.

Senior Cost Estimator/Analyst – Software SME

Must have at minimum 10 years of experience in cost analysis as well as at least seven (7) years of progressive DoW experience, e.g., Air Force, Space Force, Navy, Army,

MDA, and other DoW organizations. Preferably Air Force/ Space Force ACAT I threshold level Software program cost analysis experience.

Senior Cost Estimator/Analyst – Operations and Support SME

Must have at minimum 10 years of experience in cost analysis as well as at least seven (7) years of progressive DoW experience, e.g., Air Force, Space Force, Navy, Army, MDA, and other DoW organizations. Preferably Air Force/ Space Force ACAT I threshold level acquisition program O&S cost analysis experience.

Journeyman Cost Estimator

The contractor shall provide journeyman-level cost estimators who have experience in cost analysis principles including, but not limited to:

- a. Cost/Technical – Expertise in DAF weapon systems/system-specific cost/engineering experience to be able to identify appropriate acquisition and O&S cost drivers and document current technology issues/trends/changes driving costs.
- b. Cost Estimating – Demonstrated understanding of the connections of cost drivers and estimating the acquisition and O&S costs of weapon systems and subsystems.
- c. Data collection – Demonstrated ability to collect and normalize cost, technical, and programmatic data from Government and Industry entities. Knowledge of the government systems to which their assigned project relates, including but not limited to: Defense Cost and Resource Center (DCARC) data collection process and demonstrated experience in the use of and analysis of CSDRs and CPRs, knowledge of the DCARC data system(s) (i.e., DACIMS, CADE), knowledge of the AFTOC system.
- d. Data analysis & cost tools – Demonstrated ability to analyze data and develop cost estimating methods and tools.
- e. Experience – Must have at least five (5) years of cost analysis experience.

Junior Cost Estimator

The contractor shall provide junior-level cost estimators who are beginning to develop experience in cost analysis principles including, but not limited to:

- a. **Cost/Technical** – Some basic familiarity with DAF weapon systems, cost/engineering concepts and a desire to learn how to identify appropriate acquisition and O&S cost drivers.
- b. **Cost Estimating** – Basic understanding of cost drivers and the concepts of estimating the acquisition and O&S costs of weapon systems and subsystems.
- c. **Data collection** – Ability to collect cost, technical, and programmatic data from Government and Industry entities under supervision.
- d. **Data analysis & cost tools** – Basic ability to analyze data and a willingness to learn how to develop cost estimating methods and tools.

- e. **Experience** – No minimum required.

Expert Data Scientist

The contractor shall provide expert data scientists who possess exceptional expertise and leadership in data science, including, but not limited to:

- a. Expert Programming Skills: Mastery of Python/R with advanced skills in data manipulation libraries (e.g., Pandas, NumPy) and deep learning frameworks (e.g., TensorFlow, PyTorch), including the ability to develop custom algorithms and optimize performance.
- b. Data Pipeline Architecture: Expertise in designing, implementing, and optimizing scalable and robust data pipelines using ETL (Extract, Transform, Load) processes, distributed computing frameworks (e.g., Spark), and cloud-based data warehousing solutions. Recognized thought leader in advanced knowledge of Databricks or similar to ensure efficient data collection, transformation, and loading.
- c. Advanced Model Deployment and Optimization: Proven ability to deploy, monitor, and optimize complex machine learning models into production environments at scale, with extensive experience in A/B testing, model retraining, and performance monitoring. Demonstrated experience with MLOps principles and tools.
- d. Cutting-Edge Statistical Analysis: Mastery of statistical methods and algorithms, including advanced Bayesian methods, causal inference techniques, and time series analysis, enabling the development of innovative models and groundbreaking insights from complex, unstructured datasets.
- e. Data Storytelling and Influence: Exceptional skill in presenting data findings, actionable insights, and strategic recommendations to executive-level audiences using compelling data visualizations and narratives. Ability to influence decision-making at the highest levels of the organization.
- f. Thought Leadership and Mentorship: Recognized thought leader in the field of data science. Demonstrated ability to lead and mentor cross-functional data science teams, foster a culture of innovation, and contribute to the advancement of the field through publications, presentations, and open-source contributions.
- g. Experience: Must have at least fifteen (15) years of data science experience.

Senior Data Scientist

The contractor shall provide senior data scientists who must have experience in data science including, but not limited to:

- a. Advanced Programming Skills: Proficient in Python/R with advanced skills in data manipulation libraries (e.g., Pandas, NumPy) and machine learning frameworks (e.g., TensorFlow, scikit-learn).
- b. Data Pipeline Development: Experience designing and implementing robust data pipelines using ETL (Extract, Transform, Load) processes and frameworks.

- Advanced knowledge of Databricks or similar to ensure efficient data collection, transformation, and loading.
- c. Model Deployment and Monitoring: Proven ability to deploy machine learning models into production environments, with experience in monitoring their performance and conducting regular updates.
- d. Advanced Statistical Analysis: Deep understanding of statistical methods and algorithms, enabling the development of sophisticated models and insights from complex datasets.
- e. Data Visualization: Expert in presenting data findings and insights using visualization tools such as Tableau or PowerBI, enabling clear communication of complex ideas.
- f. Leadership and Mentorship: Ability to lead projects, mentor junior team members, and communicate complex findings to non-technical stakeholders effectively.
- g. Must have at least ten (10) years of data science experience.

Journeyman Data Scientist

Must have experience in data science including, but not limited to:

- a. Intermediate Programming Skills: Proficiency in Python/R with experience in using data manipulation libraries and machine learning frameworks to perform exploratory data analysis and model building
- b. Data Pipeline Experience: Competence in developing and maintaining data pipelines, with knowledge of ETL processes and frameworks. Advanced knowledge of Databricks or similar to ensure efficient data collection, transformation, and loading.
- c. Model Optimization: Ability to improve existing models by tuning hyperparameters, utilizing cross-validation techniques, and implementing feature engineering best practice
- d. Data Visualization: Skill in presenting data findings and insights using visualization tools such as Tableau or PowerBI, enabling clear communication of complex ideas.
- e. Cross-Functional Collaboration: Experience working with diverse teams, translating data-driven insights into actionable recommendations for stakeholders.
- f. Must have at least five (5) years of data science experience.

Junior Data Scientist

- a. Foundational Programming Skills: Basic proficiency in Python/R and familiarity with data manipulation libraries such as Pandas and NumPy, along with a willingness to learn.
- b. Data Handling: Experience in handling and cleaning datasets, and an understanding of data warehousing solutions or basic ETL processes.
- c. Model Development: Basic knowledge of machine learning concepts, with experience applying supervised and unsupervised learning techniques on smaller datasets.
- d. Collaboration: Ability to work within a team, participating in coding reviews and contributing to shared projects with guidance from senior team members.
- e. Statistical Understanding: A fundamental grasp of statistics and the ability to apply basic statistical tests and interpret their results.

f. Experience – No minimum required.

CSDR Subject Matter Expert

Highly experienced individual to lead and support Cost and Software Data (CSDR) efforts. Must have at least ten (10) years of progressively responsible experience in weapon systems acquisition and cost estimating, specifically on major DoW or Air Force ACAT I/II threshold-level programs. Your expertise should include a strong understanding of CSDR policy, planning, validation, and compliance.

Key Skills and Experience:

- a. **CSDR Policy & Guidance:** Expert knowledge of DoDI 5000.73, DoD 5000.04-M-1, and MIL-STD 881 work breakdown structures (WBS) and mapping.
- b. **DoW Cost Data:** Familiarity with various DoW cost data sources.
- c. **CADE Standard WBS:** Proven experience mapping 1921-series and FlexFile cost data to the CADE Standard WBS templates for the planning and validation process to ensure accurate cost reporting.
- d. **Contractor Cost Data Reporting (CCDR):** Expertise with 1921 series and FlexFile cost reports and applying them to the CSDR planning and validation process.
- e. **Technical Data Reporting (TDR):** Strong understanding and experience utilizing CADE Technical Data parameter lists for CSDR planning and validation.
- f. **Software Resources Data Reporting (SRDR):** Expertise with Software Resources Data Reports and applying them to the CSDR planning and validation process.
- g. **Tools Proficiency:** Working knowledge of DACIMS, cPet, PPM, and CSDR-SR systems.
- h. **Communication:** ability to present and effectively communicate complex information clearly and concisely with senior leaders.

On-Demand Subject Matter Expert (SME)

General Description: The contractor shall provide access to Subject Matter Experts (SMEs) with specialized knowledge and experience in various technical, programmatic, and operational areas, as required by the Government. These SMEs will provide short-term, focused support to address specific tasks, challenges, or knowledge gaps. The contractor shall have the capability to rapidly identify, vet, and onboard qualified SMEs within the timeframe they are needed to support task requirements, per the approved POA&M schedule.

Scope of Expertise: SME expertise may be required in, but is not limited to:

- Logistics (e.g., Supply Chain Management, Transportation, Maintenance)
- Engineering (e.g., Systems Engineering, Electrical Engineering, Aerospace Engineering, Software Engineering)
- Other Specific Weapon Systems (e.g., Communication Systems, Radar Systems, etc)
- Financial Management

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Key Positions

The government considers the following positions as key:

- Program Manager
- Chief Cost Data Scientist
- Expert Cost Estimators/Analysts
- Journeyman Data Scientists

SERVICE SUMMARY

The contractor service requirements are summarized into performance objectives that relate directly to *mission essential items*. The performance threshold briefly describes the minimum acceptable levels of service required for each requirement and will be assessed on an “Acceptable” or “Unacceptable” basis. These thresholds are critical to mission success.

Table 1: Performance Thresholds

Performance Objective	PWS Paragraph Reference	Performance Threshold	Method of Surveillance	Remedy
SS – 1 Provide qualified people at start of Task Order	0	Performance is acceptable when a) qualified personnel are on the job at contract start unless previously negotiated by the CO, and b) the contractor shall maintain 85% staffing at minimum	100% Surveillance	Performance of this objective will be annotated in the Contractor Performance Assessment Reporting System (CPARS) Rating. Failure to meet the delivery date with 75% staffed on day 1 and 100% staffed within five (5) days for Key Personnel, IAW need reference , will result in a reduction of the fixed fee amount by 20% of quarterly fixed fee amount.
SS – 2 Maintains a stable workforce (Highly Specialized Technical and Cost Analysis Expertise)	0	Less than 5% Lapse Rate across all contracted positions.	Periodic Surveillance	Performance of this objective will be annotated in the Contractor Performance Assessment Reporting System (CPARS) Rating.

Performance Objective	PWS Paragraph Reference	Performance Threshold	Method of Surveillance	Remedy
SS – 3 Replacement and/or substitute personnel	0	Performance is acceptable when a) vacancies are filled with qualified personnel within 14 days of resignation notification by incumbent personnel to the Contractor or immediately after departure of incumbent personnel or whichever is earlier unless negotiated by the CO and b) the mission is not impacted due to position vacancies or personnel qualifications	Periodic Surveillance	Performance of this objective will be annotated in the Contractor Performance Assessment Reporting System (CPARS) Rating. Failure to meet the delivery date within 5 days for Key Personnel and 8 days for all other Personnel will result in a reduction of the fixed fee amount by 15% of the quarterly fixed fee amount if standards are not met.
SS – 4 Develop and submit all required deliverables	0	Performance is acceptable when a) 95% of deliverables are met and received on time. b) Accurate information for mission critical tasks as defined in each task section.	100% Surveillance	Re-performance, CPARS, Exercise Option, Monetary Deduction which will be under the direction of the COR. See fee schedule example for monetary deductions. If the contractor has had negative trending performance, then surveillance will increase and if positive trending performance is found, then surveillance will be reduced, etc.
SS – 5 Plan of Action & Milestones (POA&M)	0	Performance is acceptable when the POA&M with the minimum requirements, IAW, PWS 0, describing the contractor's approach to satisfy all requirements is provided within 30 calendar days after contract performance begins. As well as, providing recommended changes/updates to the government at the quarterly (every three months) meetings or as needed by direction of the government.	100% Surveillance	Performance of this objective will be annotated in the CPARS rating. Failure to meet the delivery date within 10 days will result in a reduction of the fixed fee amount by 10% of quarterly fixed fee amount, if standards are not met.

DELIVERABLES

The contractor shall provide deliverables described in Table 2 in a format mutually agreed upon by the government and the contractor. All deliverables shall be in electronic format. The government and the contractor shall agree mutually upon the format for cost collection databases. Text and graphics shall be in native format compatible with MS Office and a copy in PDF.

The following enumerated deliverables are not expected to change. Due date intervals are not expected to change, but actual dates may need to be revised depending on actual contract start date.

Table 2: Task Order Deliverables

Deliverable Item	PWS Paragraph Reference	Due Date
Deliverable 1 – Signed AFCAA NDAs, and signed OSD CAPE NDAs	0	Signed NDAs must be presented to the COR NLT five (5) business days after contract performance begins and before any support is provided on this contract. Updates and new NDAs shall be signed and presented to the COR before additional or replacement contractor personnel are provided on this contract.
Deliverable 2 – Other signed Government Organization NDAs	0	As required within thirty (30) calendar days of notification of need
Deliverable 4 Status reports, which detail tasks, status and any major problems or deficiencies	0	10 th calendar day of following month
Deliverable 5 - Overall Task Plan of Action and Milestones (POA&M), including the detailed Project Plan	0, Error! Reference source not found. , and Error! Reference source not found.	Draft within thirty (30) calendar days after award. Final within sixty (60) calendar days of award. Updates quarterly or as requested. Annual update in option years.
Deliverable 6 – Task 3 and 4 Products including: Studies, Analyses Reports and Other Required Program Reports; Data Sources Quality/Availability Reports; Data Collection Reports; All source code and notebooks enabling all data pipelines , applications, data analysis, data visualizations and other related and enabling products, including documentation in collaboration tools such as GitLab, etc., documented datasets/databases, documented method sets//tools, documented Method Models, handbooks and Standard Operating Procedure guides, training materials	Error! Reference source not found.	In accordance with approved POA&M task and subtask schedules and milestones. All draft deliverables due NLT sixty (60) calendar days before end of contract, including option years. Final deliverables due NLT ten (10) business days before the end of the contract, including option years.

Deliverable Item	PWS Paragraph Reference	Due Date
Deliverable 8 – Endorsement letter to COR confirming verification and validation (V&V) by COR that all deliverables are received and stored by government for permanent record and confirming paragraph 0 requirements were completed.	Error! Reference source not found. and 0	NLT five (5) business days prior to end of contract
Deliverable 9 – Quarterly and Kickoff Meeting agendas, and supporting meeting materials/documentation, and Minutes as requested	0 and all task areas	The agenda and supporting meeting documents/briefs will be due forty-eight (48) hours prior to the meeting. Meeting minutes are due to the COR or PM within five (5) business days of the meeting.
Deliverable 10 – Travel Report		The contractor shall notify the COR or PM at least ten (10) business days prior to travel for approval. The contractor shall provide an estimated cost for the trip, description, length of days and number of travelers to the COR for approval. A Trip Report, detailing actual costs, a description of the trip, meeting minutes or action items, and any other information directed by the Government, will be submitted within five (5) business days of travel completion. In accordance with the Joint Travel Regulations (JTR), receipts are required for all lodging expenses and for any other single authorized expense of \$75.00 or more. All required receipts will be submitted with the corresponding travel invoice and referenced in the Trip Report.
Deliverable 12 – Quality Control Plan	0	Within 30 days of contract award

Status Reports

Status Reports shall include:

Status on all tasks and subtasks to include, subject to additional detail per AFCAA's request (primary intended audience being AFCAA PM and project and technical leads)

1. Task status: Open, assigned, in progress, or closed
2. Priority: Low, Med or High
3. Tracking: Delayed, On time
4. Latest progress [narrative]: list any major accomplishments since the last status report
5. Any key issues/risks with the project

Status Report will also include, from an overarching CEMDAR perspective, with intended audience as AFCAA TD, PM, COR and any other contract management personnel identified by AFCAA:

1. Schedule of planned resources, if any significant deltas to prior status report, especially with respect to new tasks opened since the prior status report

2. Identification of any major problems/deficiencies with impact, and recommended solutions.
3. Identification of any recommended/approved changes to the POA&M schedule log (reference paragraph 0.b).
4. All completed and anticipated travel.
5. Monthly expenditure log by task, and POA&M activity – actual monthly and cumulative labor hours and cost, planned monthly and cumulative labor hours and cost, funding totals, the estimate to complete, and an accurate estimate at completion.

Associate Contractor Interface

During the performance of this contract, the contractor shall be required to interface with other contractors supporting AFCAA and OSD CAPE. All such interactions will be governed by the terms of the Associate Contractor Agreement (ACA) clause, AFFARS 5352.217-9010.

To mitigate the risk of inappropriate information sharing, the contractor shall not independently share information with associate contractors. All exchanges of methods, applications, program history, or other data shall occur only at the express direction of the Contracting Officer's Representative (COR). The COR will act as the central conduit for managing communications and information sharing between associate contractors.

Under no circumstances shall the contractor represent AFCAA in policy discussions, decision-making forums, or in communications with external entities. The contractor's role is strictly limited to providing the technical and analytical services defined in this PWS.

When directed by the COR, the contractor shall perform the following actions:

1. Support the integration of the overall AFCAA team by sharing its methods, applications, and relevant program history to familiarize associate contractors with its work.
2. Provide input and analysis to the COR to assist in the development of data collection plans, including the identification of cost drivers and data gaps.
3. Notify the COR of any identified duplication of effort between its tasks and those of associate contractors.
4. Participate in integrated project teams with associate contractors. In cases where an associate contractor is designated as the technical lead for a specific work product, the COR will provide all tasking. The contractor remains responsible for the quality control of its contributions to the final delivered product.

Data Rights

The Government's rights in all data and deliverables furnished under this task order shall be governed by the applicable Defense Federal Acquisition Regulation Supplement (DFARS) clauses incorporated into this contract. The contractor shall provide all deliverables in a format

that is accessible to and fully usable by the Government, ensuring AFCAA can utilize the data for the purposes intended by this contract.

Data obtained or generated in the performance of this contract shall be maintained on secure systems, and access shall be strictly limited to contractor personnel assigned to this task.

Upon contract completion or termination, all materials produced or developed shall be turned over to the Contracting Officer's Representative (COR) in the manner described in Section 2.0. The contractor is prohibited from retaining copies of any data or materials associated with this contract beyond the period of performance and shall certify the complete destruction of all such data to the COR upon contract end.

EQUIPMENT, MATERIAL, INFORMATION, OR SERVICES

Equipment, Material, Information, or Services

Government laptop computers will be provided to those contractors as required to perform contract tasks. This will include software licenses for applications used by AFCAA in its data analytic initiatives. The Government will provide the cloud-based platform of choice as well as other associated platform requirements and licenses, for example currently tenant space(s) and licenses within the Air Force Data Visible, Accessible, Understandable, Linked and Trusted (VAULT). Access to the VAULT requires physical connection on a .mil network or through a .mil virtual private network. Government laptop computers will be provided to those contractors as required to perform contract tasks. This will include software licenses for applications used by AFCAA in its data analytic initiatives.

Contractor Acquired Property

The contractor will not be required to acquire any property.

GENERAL INFORMATION

Duty Hours

In accordance with the PWS, normal duty hours are 0800-1630, Monday through Friday. Federal holidays shall be observed in accordance with Office of Personnel Management policy. For contractual purposes, work is expected to be performed at the contractor's own facilities; however, the contractor may be requested to work on-site at government locations in the National Capital Region (NCR) in special circumstances, where it is deemed to be more efficient or effective to complete a particular activity. Work should be coordinated with the government technical lead for this effort. Contractor personnel are expected to perform during the AFCAA's normal operating hours, which are from Monday through Friday, 8:00 a.m. to 4:30 p.m.; however, these hours can be flexible at the discretion of the COR and the government technical lead. Compensatory time arrangements are permissible for temporary spikes in worktime, but overtime pay is prohibited without COR and CO approval.

Place of Performance

The place of performance for this effort is expected to be at the contractor's own facilities. The contractor's facilities do not have to be geographically located in the NCR. The contractor may make their own arrangements with employees and subcontractors to include telework and/or remote arrangements as long as contractual requirements can be met and this is at no additional cost to the government.

Kickoff Meeting/Orientation Meeting

The contractor shall schedule and coordinate a project kick-off meeting NLT ten (10) calendar days after contract performance begins at the location approved by the government. The meeting will provide an introduction between the contractor personnel and government personnel who will be involved with the contract. The meeting will provide the opportunity to discuss technical, management, and logistic issues; travel authorization; communication processes, and reporting procedures. At a minimum, the attendees shall include key contractor personnel, AFCAA stakeholders, key government representatives, and the COR. The contractor shall provide an agenda that will include, but not be limited to, the following:

- a. Introduction of personnel
- b. Overview of project tasks
- c. Review of organization (complexity)
- d. Schedule (shows major tasks, milestones, and deliverables; planned and actual start and completion dates for each)
- e. Communication plan/lines of communication overview (between both contractor and government)
- f. Travel notification and processes
- g. Security requirements (building access, badges, Common Access Cards)
- h. Invoice procedures
- i. Reporting requirements, e.g., Monthly Status Report (MSR)
- j. Points of contact
- k. Roles and responsibilities
- l. Overview of incoming transition to include process, timeframes, and status
- m. Prioritization of contractor activities
- n. Any initial deliverables
- o. Other logistic issues
- p. Quality Control Plan (QCP)
- q. Sensitivity and protection of information
- r. Additional issues of concern (Leave/back-up support)

The contractor shall provide a draft copy of the agenda NLT three (3) days after contract performance begins for review and approval by the COR prior to finalizing. The government will provide the contractor with the number of participants for the kick-off meeting and, if the meeting is not virtual, the contractor will provide sufficient copies of the presentation for all present. The kickoff meeting location will be held at the government facility or virtually and the date and time will be agreed mutually by both parties.

Quality Control.

Quality Control Plan (QCP)

The contractor shall develop and maintain an effective quality control program to ensure services are performed in accordance with this PWS, applicable laws and regulations, and best commercial practices. The contractor shall develop and implement procedures to identify, prevent, and ensure non-recurrence of defective services. The contractor's quality control program is how it assures itself that the work complies with the requirement of the contract. The QCP shall be delivered via electronic mail within 30 days of contract award. The contractor shall deliver amendments or changes to the QCP within three business days following a change to the COR and the Contracting Officer.

Contractor Manpower Reporting Application (CMRA)

The contractor shall report ALL labor hours (including subcontractor labor hours) required for performance of services provided under this contract via the secure Contractor Manpower Reporting Application (CMRA) data collection site. The contractor is required to completely fill in all required data fields at: <http://www.SAM.gov>. Reporting inputs shall be for the labor executed during the period of performance for each government fiscal year (FY), which runs 1 October through 30 September. The Unit Identification Code (UIC) for the Air Force District Washington (AFDW) is FF16M0. While inputs may be reported anytime during the FY, all data shall be reported not later than 31 October of each fiscal year. The contractor may direct questions to the CMRA help desk.

Subcontractor Input in CMRA

Prime contractors are responsible for ensuring all subcontractor data is reported. Subcontractors will not be able to enter any data into CMRA but will enter their information into a Bulk Loader spread sheet available from the AF CMRA helpdesk. Subcontractor shall fill in columns A-C and then return it to the AF CMRA helpdesk after completion. A technician team will enter the information into CMRA.

Travel and Other Direct Costs

Travel, as well as Other Direct Costs (ODC), shall be on a cost-reimbursable basis under a separately assigned CLIN.

The contractor may be required to travel to collect data or attend meetings to fulfill the requirements of this PWS. The actual number of trips and their timing are subject to change based on Government requirements and will be discussed after the period of performance begins.

All travel must be conducted in accordance with Federal Acquisition Regulation (FAR) 31.205-46, Travel Costs, and the Joint Travel Regulations (JTR). The contractor shall submit a written request for travel to the Contracting Officer's Representative (COR) for approval prior to finalizing any arrangements. All travel must be pre-approved in writing by the COR before tickets are purchased or travel commences.

969 The Government will only reimburse costs that are allowable, reasonable, and properly
970 documented. A written trip report detailing the purpose, accomplishments, and any action items
971 is required for all travel. In accordance with the JTR, receipts are required for all lodging and
972 airfare expenses, regardless of cost, and for any other individual expense of \$75.00 or more. All
973 required receipts must be submitted with the travel voucher for reimbursement.

974 Local Travel

975 All travel within 50 miles of the local city or base of assignment commuting vicinity shall be
976 considered local travel. This travel will not be separately reimbursed.

977 CONUS Travel

978 Prior to CONUS travel, the contractor shall coordinate with, and receive approval from, the COR
979 at least 10 business days prior to trip.

980 Other Direct Costs

981 At the time of this award, no specific Other Direct Costs (ODCs) are anticipated. The
982 Government expects the contractor to perform the requirements of this PWS using the resources
983 covered by the established Labor CLINs.

984 Should a future requirement for an ODC be identified, it is defined as a cost for an item essential
985 to contract performance that is not covered through another CLIN (e.g., specialized software,
986 required data purchases). The contractor shall not procure any ODC without first obtaining
987 written pre-approval from the Government.

988 If a need arises, the following Government-led approval process will be enforced:

- 989 1. Any proposed ODC with an estimated cost greater than \$10,000 must be submitted with a
990 detailed justification and approved in writing by the Contracting Officer (CO) prior to the
991 cost being incurred.
- 992 2. For proposed ODCs with an estimated cost less than or equal to \$10,000, the Contracting
993 Officer's Representative (COR) may provide written pre-approval.

994 All approved ODCs are cost-reimbursable. The contractor must submit all receipts and proof of
995 prior approval with the corresponding invoice. The Government reserves the right to deny
996 reimbursement for any ODC charge submitted without the required prior written authorization.

997 Employee Accountability and Turnover

998

999 Staffing of Contractor Personnel

1000 The contractor shall provide a stable workforce, throughout the duration of the contract, by
1001 maintaining a 10% or less employee turnover rate. No positions shall remain vacant for more
1002 than 14 business days. The government shall not be billed for positions left vacant over 14
1003 working days unless the vacancy is due to government delay and otherwise approved by the CO.

List of Employees

The contractor shall maintain a current listing of employees assigned under this contract and define who the key personnel are. The list shall include the employee's name, and level of security clearance. The list shall be validated and signed by the company's Facility Security Officer (FSO) and provided to the Contracting Officer, Program Manager, and Servicing Security Activity. An updated listing shall be provided upon request.

Contractor Professionalism

The contractor shall:

- Present a professional appearance and always maintain professional demeanor and conduct
- Conduct its work assignments IAW project schedules
- Function effectively and efficiently during extended periods of high pressure and stress
- Function as an integral member of a team of highly trained professionals responsible for the safety and security of DAF personnel and resources

Miscellaneous Paragraphs.

Reporting Requirements

The contractor shall report promptly to an appropriate authority (i.e., CO, PM, COR) any information or circumstances that they are aware of and may pose a threat to the security of DoW personnel, contractor personnel, resources and classified or unclassified defense information. Contractor employees shall be briefed on this requirement by their immediate supervisor upon initial on-base assignment.

6.8 Security Instructions

The contractor shall comply with the security instructions contained in the contract DD-254, DoW Contract Security Classification Specification.

6.8.1 Security Clearance Requirements. Performance under this PWS requires that all personnel have a minimum SECRET clearance. Performance of specific tasks or projects under this contract will require access to classified national security information and facilities up to the TOP SECRET level. The Contractor is responsible for providing all personnel who possess the requisite security clearances for task execution. The Contractor must possess, either organically or through established teaming agreements or subcontracts, the ability to provide personnel holding the required clearance at the time of contract award for SECRET clearances and at the time of task assignment for TOP SECRET. All security requirements, including clearance levels and any requirements for access to Sensitive Compartmented Information (SCI), will be specified in the DD Form 254, Contract Security Classification Specification, associated with each specific task or project. Contractor personnel may not begin work on classified tasks until their clearance and access eligibility have been verified by the Government.

6.8.2 Physical Security. The Contractor shall safeguard all Government property, documents and controlled forms provided for Contractor use and adhere to the Government property requirements contained in this contract. At the end of each workday, all Government facilities, equipment, and materials shall be secured by a Government POC. Contractors are not allowed to secure Government facilities, equipment, and materials.

6.8.3 Access Control. The Contractor shall establish and implement methods of ensuring that no building access instruments issued by the Government are lost, misplaced or used by unauthorized persons. Access codes shall not be shared with any person(s) outside the organization. The Contractor shall control access to all Government provided lock combinations to preclude unauthorized entry. The Contractor is not authorized to record lock combinations without written approval by the Government COR. Records with written combinations to authorized secure storage containers, secure storage rooms, or certified vaults, shall be marked and safeguarded at the highest classification level as the classified material maintained inside the approved containers.

6.8.4 Pass and Identification Items. The Contractor shall ensure the pass and identification items required for contract performance are obtained for employees and non-government owned vehicles.

6.8.5 Retrieving Identification Media. The Contractor shall retrieve all identification media, including vehicle passes, from employees who depart for any reason before the contract expires.

6.8.6 Traffic Laws. The Contractor and its employees shall comply with base traffic regulations.

6.8.7 Weapons, Firearms, and Ammunition. Contractor employees are prohibited from possessing weapons, firearms, or ammunition, on themselves or within their Contractor-owned vehicle or privately-owned vehicle while on the Pentagon Complex, Joint Base Anacostia-Bolling, or any other military facility.

6.8.8 Communications Security (COMSEC). Contractors may require access to COMSEC information on Air Force installations. The Contractor shall not require a COMSEC account. Access shall be controlled by the sponsoring agency. Access to COMSEC material by personnel is restricted to US citizens holding final US Government clearances. Such information is not releasable to personnel holding only reciprocal clearances. If it is determined the Contractor is required to access COMSEC information, the necessary training information and courses shall be provided by the COR. The DD Form 254 shall give further instructions on safeguarding and managing COMSEC material.

6.8.9 Contractor Identification. All Contractor personnel shall wear the Air Force issued Common Access Card at all times when away from their immediate work area so as to distinguish themselves from Government employees. When conversing with Government personnel during business meetings, over the telephone or via electronic mail, Contractor personnel shall identify themselves as a contractor to avoid situations arising where sensitive topics might be better discussed solely between Government employees. Contractors shall

1089 identify themselves on any attendance sheet or any coordination documents they may review.
1090 Electronic mail signature blocks shall identify their company affiliation. Where practicable,
1091 contractors occupying collocated space with their government program customer should identify
1092 their workspace area with their name and company affiliation or as a minimum "Contractor"
1093 after name.

1094
1095 6.8.10 Drug, Tobacco, and Alcohol Use Policy. The consumption of alcoholic beverages or
1096 illegal drugs by contractor personnel, while on duty, is strictly forbidden. Contractor shall
1097 immediately remove any employee who is under the influence of alcohol or drugs.

1098

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1099 Acronyms

ACAT	Acquisition Category
AF	Air Force
AFCAA	Air Force Cost Analysis Agency
AFDW	Air Force District Washington
AFI	Air Force Instruction
AFTOC	Air Force Total Ownership Cost
AIS	Automated Information System
ANSI	American National Standards Institute
AS	Acquisition Strategy
C&A	Certification and Accreditation
C4I	Command, Control, Communications, Computers, & Intelligence
CAC	Common Access Card
CADE	Cost Assessment Data Enterprise
CAPE	Cost Assessment and Program Evaluation
CARD	Cost Analysis Requirements Description
CEM	Cost Estimating Modeling
CER	Cost Estimating Relationship
CLIN	Contract List Item Number
CMRA	Contractor Manpower Reporting Application
CO	Contracting Officer
COMSEC	Communications Security
COR	Contracting Officer's Representative
COTS	Commercial-Off-The-Shelf
CPR	Contract Performance Report
CSDR	Cost and Software Data Reporting
CTO	Certificate To Operate
DACIMS	Defense Automated Cost Information Management System
DCARC	Defense Cost and Resource Center

DFARS	Defense Federal Acquisition Regulation Supplement
DIACAP	Defense Information Assurance Certification and Accreditation
DoDD	Department of Defense Directive
DoDI	Department of Defense Instruction
DoDM	Department of Defense Manual
DoW	Department of War
DSAM	Datasets, Analysis and Methods
EITSD	Enterprise IT Services Directorate
ERP	Enterprise Resource Planning
EVMIG	Earned Value Management Implementation Guide
FAR	Federal Acquisition Regulation
FOIA	Freedom of Information Act
FOUO	For Official Use Only
FSO	Facility Security Officer
FY	Fiscal Year
HQDA ASA	Headquarters, Department of the Army – Assistant Secretary of the Army
IA	Information Assurance
ICBM	Intercontinental Ballistic Missiles
ICE	Independent Cost Estimate
IPMR	Integrated Program Management Report
IT	Information Technology
JTR	Joint Travel Regulations
KSA	Knowledge, Skills, and Abilities
MDA	Missile Defense Agency
MOA	Memoranda of Agreement
NACI	National Agency Check with Inquiries
NCR	National Capital Region
NDA	Nondisclosure Agreements

NLT	No Later Than
O&S	Operating and Support
OCI	Organizational Conflict of Interest
ODC	Other Direct Costs
OEM	Original Equipment Manufacturer
OGC	Other Government Costs
OSD	Office of the Secretary of Defense
PM	Program Manager
POA&M	Plan of Action and Milestones
PROPIN	Proprietary Information
PWS	Performance Work Statement
SAR	Selected Acquisition Report
SER	Schedule Estimating Relationship
SOP	Standard Operating Procedure
SRDR	Software Resources Data Report
SSA	Servicing Security Activity
TD	Technical Director
TL	Technical Lead
TO	Task Order
UAV	Unmanned Air Vehicles
UIC	Unit Identification Code
URL	Uniform Resource Locator
USC	United States Code
VAULT	Visible, Accessible, Understandable, Linked and Trusted
VGSA	Visitor Group Security Agreement

Applicable Publications

Publications applicable to this PWS are listed below. DoW issuances are available at <https://www.esd.whs.mil/dd/>. All other publications listed are available via the Internet at the specified URL, also known as the web address.

Mandatory – Department of Defense (DoD) Directive (DoDD) 5000.01, “The Defense Acquisition System”, August 31, 2018

Mandatory – DoD Instruction (DoDI) 5000.75, “Business Systems Requirements and Acquisition,” January 24, 2020

Mandatory – DoDI 8510.01, “Risk Management Framework (RMF) for DoD Information Technology (IT)”, July 28, 2017

Mandatory – DoD Manual (DoDM) 5200.01 Volume 1, “DoD Information Security Program: Overview, Classification, And Declassification”, May 14, 2019

Mandatory – DoDM 5200.01 Volume 2 “DoD Information Security Program: Marking of Information”, May 4, 2018

Mandatory – DoDM 5200.01 Volume 3, “DoD Information Security Program: Protection of Classified Information”, March 19, 2013

Mandatory – DoDM 5400.7, “DoD Freedom of Information Act (FOIA) Program”, January 25, 2017

Mandatory – DoDI 8580.1, “Information Assurance (IA) in the Defense Acquisition System”, July 9, 2004

Mandatory – DoD 8570.01-M “Information Assurance Workforce Improvement”, November 10, 2015

Mandatory – CJCSI 6510.01F, “Information Assurance (IA) and Support to Computer Network Defense (CND)”, June 9, 2015

Mandatory – Subchapter III of Chapter 35 of title 44, United States Code, “Federal Information Security Management Act (FISMA) of 2002”

Mandatory –Memorandum of understanding Between CAPE, DCARC, and EITSD [current version]

Advisory – Defense Federal Acquisition Regulation Supplement (DFARS) Subpart 234.2

Advisory – Earned Value Management Implementation Guide (EVMIG)

Advisory – American National Standards Institute/Electronic Industries Alliance Standard 748 (ANSI/EIA-748)

Mandatory – DoDI 5000.02T, “Operation of the Defense Acquisition System”, April 21, 2020

Mandatory – DoDI 5000.02, “Operation of the Adaptive Acquisition Framework”, January 23, 2020

Mandatory – DoDI 5000.73, “Cost Analysis Guidance and Procedures”, March 13, 2020, establishes the policies for cost data collection in the DoD. The CSDR requirements and reporting thresholds are listed in Section 4: Data Collection, Table 1. Cost Data Reporting Requirements.

Mandatory – DoD 5000.4-M, “Cost Analysis Guidance and Procedures,” April 18, 2018

Mandatory - Military Standard 881D, “Work Breakdown Structures for Defense Materiel Items,” June 19, 2018

Mandatory - DD Form 2794, “Cost and Software Data Reporting Plan,” Commonly referred to as the CSDR Plan, a completed DD Form 2794 must be approved by the Office of the Secretary of Defense (OSD) Deputy Director, Cost Assessment (DDCA).

Mandatory - DD Form 1921, “Cost Data Summary Report,” available at <https://cade.osd.mil/policy/dids>

Mandatory - DD Form 1921-1, “Functional Cost-Hour Report,” available at <https://cade.osd.mil/policy/dids>

Mandatory - DD Form 1921-2, “Progress Curve Report,” available at <https://cade.osd.mil/policy/dids>

Mandatory - DD Form 1921-3, “Contractor Business Data Report,” available at <https://cade.osd.mil/policy/dids>

Mandatory - DD Form 1921-5, “Sustainment Functional Cost-Hour Report,” available at: <https://cade.osd.mil/policy/dids>

DD Form 3026-1, “Software Development Report,” available at: <https://cade.osd.mil/policy/dids>

DD Form 3026-2, “Software Maintenance Report,” available at: <https://cade.osd.mil/policy/dids>

DD Form 3026-3, “Enterprise Resource Planning (ERP) Software Development Report,” available at: <https://cade.osd.mil/policy/dids>

Data Item Description DI-FNCL-81565C, “Cost Data Summary Report (DD Form 1921),” available at <https://cade.osd.mil/policy/dids>

Data Item Description DI-MGMT-81334D, “Contract Work Breakdown Structure,” available at <https://cade.osd.mil/policy/dids>

Data Item Description DI-FNCL-81566C, “Functional Cost-Hour Report (DD Form 1921-1),” available at <https://cade.osd.mil/policy/dids>

Data Item Description DI-FNCL-81567C, “Progress Curve Report (DD Form 1921-2),” available at <https://cade.osd.mil/policy/dids>

Data Item Description DI-FNCL-81765B, “Contractor Business Data Report (DD Form 1921-3),” available at <https://cade.osd.mil/policy/dids>

Data Item Description DI-FNCL-81992, “Sustainment Functional Cost-Hour Report (DD Form 1921-5):” available at <https://cade.osd.mil/policy/dids>

Data Item Description DI-MGMT-82035A, “Software Resources Data Reporting: Development, Maintenance and Enterprise Resource Planning Development Reports, and Data Dictionary,” available at <https://cade.osd.mil/policy/dids>

Data Item Description DI-FNCL-82162, “Cost and Hour Report (FlexFile),” available at <https://cade.osd.mil/policy/dids>

Data Item Description DI-MGMT-82163, “Maintenance and Repair Parts Data Report,” available at <https://cade.osd.mil/policy/did>

Data Item Description DI-MGMT-82164, “Quantity Data Report,” available at <https://cade.osd.mil/policy/did>

Data Item Description DI-MGMT-82165, “Technical Data Report,” available at <https://cade.osd.mil/policy/did>

DFARS 252.234-7001, “Notice of Earned Value Management System” DFARS 252.234-7002, “Earned Value Management System”

DFARS 252.234-7003, “Notice of Cost and Software Data Reporting System” DFARS 252.234-7004, “Cost and Software Data Reporting System”

Air Force Instruction 16-1404, “Air Force Information Security Program,” May 29, 2015

Air Force Instruction AFI65-508, “Cost Analysis Guidance and Procedures,” December 6, 2018

Notional Plan of Action and Milestones

See separated attachment for the notional POA&M.

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Metadata/Key Word Example

Each project deliverable (draft final and final) shall include a listing of metadata that includes the following data fields – title, project lead’s organization, point of contact or project lead’s name, publication date, the report dates (from – to), container type, the life-cycle phase, description or abstract, list of all proprietary data, document accessibility, document marking and distribution statements, sponsoring organization, sponsoring service, the contract name, the contract number, programs analyzed, source data and its data fidelity (0), and a list of key terms. The specific metadata will be adjusted depending on the disposition of the deliverable, that is, whether it is uploaded to the DTM Hub or the DAF e-Library. AFCAA shall make available to the contractor templates to capture the required metadata for each deliverable (draft final and final) and the AFCAA “roll out” presentation.

AFCAA also shall make available a guide to annotate project deliverables for Controlled Unclassified Information (CUI). All proprietary data in project deliverables shall be listed. Below is an illustration of the required data that supports a Proprietary Information (PROPIN) marking.

Short Name	Component	Product	Contract	Vendor (Submitting Contractor)	Raw Data Type	Data is proprietary to?
AH-64E New Build	Army	Airframe	XXXXXX-XX-X-0001	Boeing	CCDR, SRDR, CWBS, IPMR	Boeing
AH-64E New Build	Army	Avionics	XXXXXX-XX-X-0002	Honeywell	CCDR, SRDR, CWBS	Honeywell
AH-64E New Build	Army	TADS/PNV S	XXXXXX-XX-X-0003	Lockheed Martin	CCDR, SRDR, CWBS	Lockheed Martin
AH-64E Remanufacture	Army	Airframe	XXXXXX-XX-X-0001	Boeing	Ktr site visit accounting data, Bill of materials, CDR Briefing	Boeing
AH-64E Remanufacture	Army	Engine	XXXXXX-XX-X-0002	Pratt & Whitney	CCDR, SRDR, CWBS	Pratt & Whitney
UH-60M Black Hawk	Army	Airframe	XXXXXX-XX-X-0001	Lockheed Martin Sikorsky	CCDR, SRDR, CWBS, IPMR	Lockheed Martin Sikorsky
UH-60M Black Hawk	Army	Avionics	XXXXXX-XX-X-0002	Honeywell	CCDR, SRDR, CWBS	Honeywell
CH-53K	Navy	Airframe	XXXXXX-XX-X-0001	Lockheed Martin Sikorsky	CCDR, SRDR, CWBS, PDR &	Lockheed Martin Sikorsky

Short Name	Component	Product	Contract	Vendor (Submitting Contractor)	Raw Data Type	Data is proprietary to?
					CDR Briefings	

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Data Fidelity Scorecard Example

Authenticity Assessment Factor		Completeness Assessment Factor 1		Completeness Assessment Factor 2		Color Rating	Assessed Score
Data is from a verifiable official source provided to the Government (e.g., CSDR, Flex File, IPMR, other CDRLs)	AND	Percent complete is 95% or greater {If not see lower rating categories below to determine rating}	IF YES →	Data is raw or in a form that allows for renormalization (both inflation/escalation and relevant mapping/allocation to standard WBS e.g. 881D/CSDR standard plans). If not, see below categories to make final determination}.	THEN		7
IF NO ↓		IF NO ↓					
Data is from a recognized data capture system (e.g. informal data from contractor accounting system) or generated by the government for official reporting purposes (e.g. SAR, Budget Doc, etc.)	AND	Percent complete is 85% or greater {If not see lower rating categories below to determine rating}	IF YES →	1. Data is raw or in a form that allows for renormalization (both inflation/escalation and relevant mapping/allocation) OR 2. Data is in a normalized form and has documentation that allows for a high level of renormalization (inflation/escalation renormalization ability and some degree of relevant mapping/allocation). If not 1 or 2, see Yellow and Red categories below	THEN		6= If Authenticity was blue OR 5 = If Authenticity was Green
IF NO ↓		IF NO ↓					

Authenticity Assessment Factor		Completeness Assessment Factor 1		Completeness Assessment Factor 2		Color Rating	Assessed Score
Data is from a reliable study, organization, or a systematic data capture process. Includes significant amount of the data that represents contract price paid rather than actual costs incurred (e.g. FFP values that don't represent true cost of the effort or product)	AND	Percent complete is 70% or greater {If not, rating is red, see below}	IF YES →	1. Data is raw or in a form that allows for renormalization (both inflation/escalation and relevant mapping/allocations) OR 2. Data is in a normalized form and has documentation that allows for a high level of renormalization (inflation/escalation renormalization ability and some degree of relevant mapping/allocations)? OR 3. Data is only in a normalized form, but documentation provides a limited understanding of how it was normalized and/or limited traceability to do renormalization. If not 1, 2. or 3, then see red category below	THEN		4 = If authenticity was blue or green OR 3 = If authenticity was yellow
IF NO ↓				IF NO ↓			

Authenticity Assessment Factor		Completeness Assessment Factor 1		Completeness Assessment Factor 2		Color Rating	Assessed Score
Data is from a 3rd party data source (e.g. public website, limited documented study/briefing/report with little traceability to source/reliability of the data) or a contractor estimate from a proposal or data is anecdotal.	OR	Percent complete is <u>70% or lower</u>	OR	Data is not raw data, and no insight exists into inflation/escalation normalization or exhibits significant lack of insight into mapping/allocations	THEN		2= If authenticity was blue, yellow, or green OR 1 = If authenticity was red

APPENDIX A: Example List of Projects

Project 1: Aircraft Development Expenditure Time Phasing Database/Analysis

The study objective is to improve time phasing of development cost estimates. The process will include collecting, organizing, normalizing, and analyzing time phased expenditure data from aircraft programs (fixed wing, rotary wing, UAV, and associated subsystems). We will obtain data from OSD Earned Value Central Repository (IPMR, CFSR), CADE, and other government Sources. We will assess and potentially improve upon the initial database structure to house the data, populate the database (establish WBS level of detail with sponsor), perform analysis and recommend methods or approaches for time phasing future cost estimates. Demonstrate this as a use case for the EVM Data Layer.

Project 2: Aircraft Schedule Dataset/Database, Analysis, and SERs

The study objective is to continue improving development schedule estimating through additional data capture and analysis. We will review data sources, assess and update the existing AFCAA schedule dataset as new information is available. Planned deep dive analysis focuses on Commercial Derivative, Fighter, Tanker, and Trainer programs.

Project 3: Engine Database/Benchmarks/Methods

The study objective is to improve cost estimating capabilities for engines but developing an organized set of cost, technical and programmatic engine data and benchmarks. We will collect/organize cost, technical, and programmatic data for engine cost elements into a user-friendly dataset for use by cost analysts. We will conduct analysis and develop benchmarks and/or methods for cost estimating.

Project 4: Aircraft Test Cost/Execution Database and Benchmarks

The study objective is to develop an all-encompassing database to capture government test costs by fiscal year and by cost category, contractor test hours/costs, and available test execution data (# aircraft, # flights/flight hours, schedule, etc.) to allow development of test metrics/benchmarks for use in cost estimating.

Project 5: Drivers to Underestimating Aircraft Integration for Modification and Derivative Programs

The study's objective is to identify causes for underestimating cost, schedule, and technical complexity for modification and derivative programs. We intend to collect, organize, normalize, analyze and compare early cost estimates with later actual cost data from aircraft programs that incorporated a new system into an existing platform. In addition, we will compare and contrast the early technical complexity assumptions of the integration effort with the actual technical complexity. Throughout the study we will obtain

data from identified sources, recommend a database structure to house the data, populate the database (establish WBS level of detail with sponsor), perform analysis, and document findings of the research.

Project 6: Aircraft Install Dataset

The study objective is to develop a dataset to capture government installation hours/costs and available ALC (Aircraft Logistics Complex) installation data. This will allow the development of installation metrics/benchmarks for use in cost estimating especially for modification efforts.

Project 7: Contracts Database, Benchmarks, and Deep-dives

The study objective update an existing AFCAA Contracts Database with additional contract information for programs that already exist in the database as well as add new programs to the database. AFCAA will identify new priorities for the Contracts Database at the start of each option period. Additionally, the study will update an existing AFCAA Improving Cost Estimating Approaches Analysis.

Project 1: Funding Impact on Program Schedules

Analyze how annual funding fluctuations affect space program cost and schedule. This involves collecting and normalizing funding data against program cost estimates and key delivery dates to determine the relationship between budget changes and program delays.

Project 2: Commercial-Like Cost Estimation

Develop guidelines for applying "commercial-like" adjustments to Government space program cost estimates. This will be achieved by researching commercial satellite industry manufacturing and integration processes and using that data to create rules of thumb or Cost Estimating Relationships (CERs) for government programs.

Project 3: Technical Parameter Standardization

Identify core sets of technical parameters for data collection across different space commodities. The goal is to streamline the collection of useful technical data by examining the key parameters that drive cost-estimating methods for various space agencies. This will result in guidance for standardizing data collection for future analysis.